

Machine Learning

Homework 1

Oct. 2025

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1. In linear regression, we learned to find the optimal parameters by minimizing the loss function. However, when the model is too complex or the feature dimension is too high, overfitting is prone to occur. Regularization is a key technique to solve this problem.

- (a) Briefly explain the core differences between Ridge Regression and Lasso Regression regarding the following two aspects:

1. Their effect on model weights.
2. Their suitability for feature selection. Which method is more suitable and why?

(20pts)

- (b) Consider the cost function for Ridge Regression:

$$J(\mathbf{w}, b) = \frac{1}{m} \sum_{i=1}^m (\mathbf{w}^T \mathbf{x}_i + b - y_i)^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2, \quad (1)$$

Let $\tilde{\mathbf{w}} = \begin{bmatrix} \mathbf{w} \\ b \end{bmatrix}$ and $\tilde{\mathbf{x}}_i = \begin{bmatrix} \mathbf{x}_i \\ 1 \end{bmatrix}$ to absorb the bias term into the weight vector.

Rewrite the cost function in its matrix form, $J(\tilde{\mathbf{w}})$. Then, derive the closed-form solution for $\tilde{\mathbf{w}}$ by solving $\nabla_{\tilde{\mathbf{w}}} J(\tilde{\mathbf{w}}) = 0$. Finally, prove that this solution is unique for any regularization parameter $\lambda > 0$.

(10,10,10pts)

Ans:

- (a) The difference between Ridge Regression and Lasso Regression lies in two aspects (bold text are important for points):

1. The term in Ridge Regression is $\lambda \|\mathbf{w}\|_2^2 = \lambda \sum w_j^2$. It penalizes large weights, **tending to shrink all weights close to zero, but not exactly zero** (unless the weights themselves are very small). It makes the weight distribution smoother. The term in Lasso Regression is

$\lambda \|\mathbf{w}\|_1 = \lambda \sum |w_j|$. Due to the geometric shape of the L1 norm, during the optimization process, the contours of the loss function are more likely to intersect with the vertices of the L1 norm, which **causes some weights to be compressed exactly to zero**.

2. **Lasso is more suitable** because it **produces a sparse solution**, meaning many model coefficients are zero. Features with zero coefficients are effectively eliminated by the model, thus achieving automatic feature selection.

(b) Let the matrix be $\tilde{\mathbf{X}} = \begin{bmatrix} \mathbf{x}_1^T & 1 \\ \mathbf{x}_2^T & 1 \\ \vdots & \vdots \\ \mathbf{x}_m^T & 1 \end{bmatrix}_{m \times (d+1)}$, the target vector be $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}_{m \times 1}$.

(Just as the form like Lec2. page 15) The loss function for Ridge Regression can be rewritten as

$$J(\tilde{\mathbf{w}}) = \frac{1}{m}(\mathbf{y} - \tilde{\mathbf{X}}\tilde{\mathbf{w}})^T(\mathbf{y} - \tilde{\mathbf{X}}\tilde{\mathbf{w}}) + \lambda \mathbf{w}^T \mathbf{w}.$$

(10 pts for the right form.)

To fit the derivative calculation, let's set $\mathbf{L} = \begin{bmatrix} \mathbf{I}_{d \times d} & 0 \\ 0 & 0 \end{bmatrix}_{(d+1) \times (d+1)}$, where $\mathbf{I}_{d \times d}$ is a $(d+1)$ -dimension matrix, then the above form can be rewritten as

$$J(\tilde{\mathbf{w}}) = \frac{1}{m}(\mathbf{y} - \tilde{\mathbf{X}}\tilde{\mathbf{w}})^T(\mathbf{y} - \tilde{\mathbf{X}}\tilde{\mathbf{w}}) + \lambda \mathbf{w}^T \mathbf{L} \mathbf{w}.$$

The derivative for the loss function is

$$\nabla_{\tilde{\mathbf{w}}} J(\tilde{\mathbf{w}}) = \frac{2}{m} \tilde{\mathbf{X}}^T (\tilde{\mathbf{X}}\tilde{\mathbf{w}} - \mathbf{y}) + 2\lambda \mathbf{L}\tilde{\mathbf{w}}.$$

(6 pts for the right form.)

We set $\nabla_{\tilde{\mathbf{w}}} J(\tilde{\mathbf{w}}) = 0$, then

$$\begin{aligned} \frac{2}{m} \tilde{\mathbf{X}}^T (\tilde{\mathbf{X}}\tilde{\mathbf{w}} - \mathbf{y}) + 2\lambda \mathbf{L}\tilde{\mathbf{w}} &= 0 \\ (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + m\lambda \mathbf{L})\tilde{\mathbf{w}} &= \tilde{\mathbf{X}}^T \mathbf{y} \\ \tilde{\mathbf{w}}^* &= (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + m\lambda \mathbf{L})^{-1} \tilde{\mathbf{X}}^T \mathbf{y} \end{aligned}$$

(4pts for the right form.)

Next we give a proof of the solution is unique for any regularization parameter $\lambda > 0$. (10 pts for a reasonable proof.)

Proof:

The solution is unique if and only if $(\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + m\lambda \mathbf{L})$ is invertible. We can try to prove $(\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + m\lambda \mathbf{L})$ is a positive-definite matrix and then the matrix is also invertible. We prove this using the definition of positive-definite matrix.

For any non-zero vector $\mathbf{v} \in \mathbb{R}^{d+1}$, we have

$$\mathbf{v}^T (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + m\lambda \mathbf{L}) \mathbf{v} = \mathbf{v}^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \mathbf{v} + m\lambda \mathbf{v}^T \mathbf{L} \mathbf{v} = \|\tilde{\mathbf{X}} \mathbf{v}\|_2^2 + m\lambda \mathbf{v}^T \mathbf{L} \mathbf{v}.$$

For $\mathbf{v}^T \mathbf{L} \mathbf{v} = \sum_{i=1}^d v_i^2$, where v_i^2 is the i -th element of $\mathbf{v}$, we have

- If at least one of the top d -th element of $\mathbf{v}$ is not equal to 0, then $\mathbf{v}^T \mathbf{L} \mathbf{v} > 0$.
- If all of the top d -th element of $\mathbf{v}$ is 0, then $\mathbf{v} = [0, \dots, 0, c]^T$, where $c \neq 0$. We got $\tilde{\mathbf{X}} \mathbf{v} = [c, \dots, c]^T \in \mathbb{R}^m$, so $\|\tilde{\mathbf{X}} \mathbf{v}\|_2^2 > 0$.

In summary, for any non-zero vector $\mathbf{v} \in \mathbb{R}^{d+1}$ we have $\mathbf{v}^T (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + m\lambda \mathbf{L}) \mathbf{v} > 0$. According to the definition of positive-definite matrix, $\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + m\lambda \mathbf{L} > 0$, then it is a invertible matrix, which implies that the solution is unique. $\square$

2. Consider the binary cross-entropy loss for a single training example $(\mathbf{x}_i, y_i)$:

$$J_i(\boldsymbol{\beta}) = -[y_i \log(p_1(\hat{\mathbf{x}}_i; \boldsymbol{\beta})) + (1 - y_i) \log(1 - p_1(\hat{\mathbf{x}}_i; \boldsymbol{\beta}))], \quad (2)$$

where $\boldsymbol{\beta} = (\mathbf{w}; b)$, $\hat{\mathbf{x}}_i = (\mathbf{x}_i; 1)$, $p_1(\hat{\mathbf{x}}_i; \boldsymbol{\beta}) = \sigma(\boldsymbol{\beta}^T \hat{\mathbf{x}}_i)$ is the predicted probability that $y_i = 1$, and $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.

- Derive the gradient of this loss function with respect to the parameter vector $\boldsymbol{\beta}$, which is $\nabla_{\boldsymbol{\beta}} J_i(\boldsymbol{\beta})$. (Hint: You will need to use the property of the sigmoid function's derivative: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.) (20pts)
- Now, let's add an L2 regularization term to the logistic regression cost function. The new cost function for the entire dataset is:

$$J_{\text{reg}}(\boldsymbol{\beta}) = \left(\frac{1}{m} \sum_{i=1}^m J_i(\boldsymbol{\beta}) \right) + \frac{\lambda}{2m} \|\mathbf{w}\|_2^2. \quad (3)$$

Note that regularization is applied only to the weights $\mathbf{w}$, not the bias term b . Write down the gradient descent update rules for both the weights $\mathbf{w}$ and the bias b based on this regularized cost function $J_{\text{reg}}(\boldsymbol{\beta})$. (20pts)

Ans:

(a) For simplicity, let's set $p_1 = p_1(\hat{\mathbf{x}}_i; \boldsymbol{\beta})$, $\mathbf{z} = \boldsymbol{\beta}^T \hat{\mathbf{x}}_i$, then the loss function is

$$J_i(\boldsymbol{\beta}) = -[y_i \log(p_1) + (1 - y_i) \log(1 - p_1)].$$

By applying the chain rule of derivative, we got $\nabla_{\boldsymbol{\beta}} J_i(\boldsymbol{\beta}) = \frac{\partial J_i}{\partial p_1} \frac{\partial p_1}{\partial \mathbf{z}} \frac{\partial \mathbf{z}}{\partial \boldsymbol{\beta}}$.

$$\begin{aligned} \frac{\partial J_i}{\partial p_1} &= -\frac{y_i}{p_1} + \frac{1 - y_i}{1 - p_1} = \frac{-y_i(1 - p_1) + p_1(1 - y_i)}{p_1(1 - p_1)} = \frac{p_1 - y_i}{p_1(1 - p_1)}, \\ \frac{\partial p_1}{\partial \mathbf{z}} &= \sigma'(\mathbf{z}) = \sigma(\mathbf{z})(1 - \sigma(\mathbf{z})) = p_1(1 - p_1), \\ \frac{\partial \mathbf{z}}{\partial \boldsymbol{\beta}} &= \frac{\partial(\boldsymbol{\beta}^T \hat{\mathbf{x}}_i)}{\partial \boldsymbol{\beta}} = \hat{\mathbf{x}}_i. \end{aligned}$$

By substituting those equations into $\nabla_{\boldsymbol{\beta}} J_i(\boldsymbol{\beta})$, we got

$$\begin{aligned} \nabla_{\boldsymbol{\beta}} J_i(\boldsymbol{\beta}) &= \frac{\partial J_i}{\partial p_1} \frac{\partial p_1}{\partial \mathbf{z}} \frac{\partial \mathbf{z}}{\partial \boldsymbol{\beta}} \\ &= \frac{p_1 - y_i}{p_1(1 - p_1)} p_1(1 - p_1) \hat{\mathbf{x}}_i \\ &= (p_1 - y_i) \hat{\mathbf{x}}_i \\ &= (\sigma(\boldsymbol{\beta}^T \hat{\mathbf{x}}_i) - y_i) \hat{\mathbf{x}}_i. \end{aligned}$$

(b) For the partial derivative of J_{reg} with respect to $\mathbf{w}$, we have

$$\begin{aligned} \frac{\partial J_{\text{reg}}}{\partial \mathbf{w}} &= \frac{1}{m} \sum_{i=1}^m \frac{\partial J_i}{\partial \mathbf{w}} + \frac{\lambda}{m} \mathbf{w} \\ &= \frac{1}{m} \sum_{i=1}^m (\sigma(\boldsymbol{\beta}^T \hat{\mathbf{x}}_i) - y_i) \hat{\mathbf{x}}_i + \frac{\lambda}{m} \mathbf{w}. \end{aligned}$$

For the partial derivative of J_{reg} with respect to b , we have

$$\begin{aligned} \frac{\partial J_{\text{reg}}}{\partial b} &= \frac{1}{m} \sum_{i=1}^m \frac{\partial J_i}{\partial b} \\ &= \frac{1}{m} \sum_{i=1}^m (\sigma(\boldsymbol{\beta}^T \hat{\mathbf{x}}_i) - y_i) \cdot 1. \end{aligned}$$

Thus, the gradient descent update rules for both the weights $\mathbf{w}$ and the bias b are

$$\begin{aligned} \mathbf{w} &\leftarrow \mathbf{w} - \eta \left(\frac{1}{m} \sum_{i=1}^m (\sigma(\boldsymbol{\beta}^T \hat{\mathbf{x}}_i) - y_i) \hat{\mathbf{x}}_i + \frac{\lambda}{m} \mathbf{w} \right), \\ b &\leftarrow b - \eta \left(\frac{1}{m} \sum_{i=1}^m (\sigma(\boldsymbol{\beta}^T \hat{\mathbf{x}}_i) - y_i) \right), \end{aligned}$$

where η is the learning rate.

3. For the standard multivariate linear regression model, the model is expressed in matrix form as $\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}$. The ordinary least squares solution, which minimizes the residual sum of squares $\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$, is given by the normal equation: $\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ (assuming $\mathbf{X}^T\mathbf{X}$ is invertible). The predicted values are $\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}^*$, and the residual vector is defined as $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$.

Prove that the residual vector $\mathbf{e}$ is orthogonal to the column space of the feature matrix $\mathbf{X}$. Mathematically, this means proving that $\mathbf{X}^T\mathbf{e} = \mathbf{0}$. (10pts)

Ans: Proof:

By substituting $\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ into $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\mathbf{w}^*$, we have $\mathbf{e} = \mathbf{y} - \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$, then

$$\begin{aligned}\mathbf{X}^T\mathbf{e} &= \mathbf{X}^T(\mathbf{y} - \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}) \\ &= \mathbf{X}^T\mathbf{y} - \mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \\ &= \mathbf{X}^T\mathbf{y} - \mathbf{I}\mathbf{X}^T\mathbf{y} \\ &= \mathbf{0},\end{aligned}\tag{4}$$

where $\mathbf{I}$ is the identity matrix. □